

Lecture 9

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Midterm Exam Info

The midterm exam will cover material up through Lecture 9, but not including of Convex Sets and Convex Functions. This includes:

- Function properties: differentiable, continuous, Lipschitz, etc, in $\mathbb{R}$ as well as $\mathbb{R}^n$, and $\mathbb{R}^{m \times n}$
- Analysis of convergence rates
- Newton's Method and variants
- Vector spaces and inner products
- Derivatives, gradients, Jacobian, Hessian. Definitions, computing them in different scenarios, with chain rule etc.
- Definite and semidefinite matrices and their properties
- Multivariate Taylor expansion

Difficulty

The problems will be somewhat more straightforward than your homework assignments. They will involve computing some things (derivatives, convergence rates), some simple proofs, and some conceptual questions.

Good practice would be working through the exercises and examples in the lectures.

Notes

You may bring a single-sided page of notes.

If you receive accommodations or know you will be absent, please reach out to me as soon as possible. If you are ill on the day of the exam, we will be able to provide a makeup, please email me.

Linear and quadratic approximations and Taylor's theorem

Reinterpreting gradients as linear approximations

For a C^1 function $f : \Omega \rightarrow \mathbb{R}$, its gradient is defined by

$$\lim_{\mathbf{h} \rightarrow 0} \frac{|f(\mathbf{x} + \mathbf{h}) - f(\mathbf{x}) - \nabla f(\mathbf{x})^\top \mathbf{h}|}{\|\mathbf{h}\|} = 0$$

if f is vector-variate and

$$\lim_{H \rightarrow 0} \frac{|f(X + H) - f(X) - \text{tr}(\nabla f(X)^\top H)|}{\|H\|} = 0$$

if f is matrix-variate.

These may be reinterpreted as a first order or linear approximation

$$\begin{aligned} f(\mathbf{x} + \mathbf{h}) &\approx f(\mathbf{x}) + \nabla f(\mathbf{x})^\top \mathbf{h}, \\ f(X + H) &\approx f(X) + \text{tr}(\nabla f(X)^\top H) \end{aligned}$$

respectively.

For a C^2 function $f : \Omega \rightarrow \mathbb{R}$, its gradient and Hessian may be simultaneously defined via

$$\lim_{\mathbf{h} \rightarrow 0} \frac{|f(\mathbf{x} + \mathbf{h}) - f(\mathbf{x}) - \nabla f(\mathbf{x})^\top \mathbf{h} - \frac{1}{2} \mathbf{h}^\top \nabla^2 f(\mathbf{x}) \mathbf{h}|}{\|\mathbf{h}\|^2} = 0$$

if f is vector-variate and

$$\lim_{H \rightarrow 0} \frac{|f(X + H) - f(X) - \text{tr}(\nabla f(X)^\top H) - \frac{1}{2} \text{tr}(H^\top [\nabla^2 f(X)] (H))|}{\|H\|^2} = 0.$$

if f is matrix-variate.

These may be reinterpreted as a second order or quadratic approximation

$$f(\mathbf{x} + \mathbf{h}) \approx f(\mathbf{x}) + \nabla f(\mathbf{x})^\top \mathbf{h} + \frac{1}{2} \mathbf{h}^\top \nabla^2 f(\mathbf{x}) \mathbf{h},$$

$$f(X + H) \approx f(X) + \text{tr}(\nabla f(X)^\top H) + \frac{1}{2} \text{tr}(H^\top [\nabla^2 f(X)] (H))$$

respectively.

For example, by our previous calculations, the quadratic approximation of the $\log \det$ function on $\mathbb{S}_{++}^n$ is

$$\log \det(X + H) \approx \log \det(X) + \operatorname{tr}(X^{-1}H) - \frac{1}{2} \operatorname{tr}(HX^{-1}HX^{-1}),$$

and the quadratic approximation of the trace inverse function on $\mathbb{S}_{++}^n$ is

$$\operatorname{tr}(X + H)^{-1} \approx \operatorname{tr}(X^{-1}) - \operatorname{tr}(X^{-2}H) + \frac{1}{2} \operatorname{tr}(HX^{-1}HX^{-2} + HX^{-2}HX^{-1}).$$

Taylor's theorem may be viewed as a rigorous version of the linear and quadratic approximations. It is also a fundamental tool in multivariate optimization.

Slight change in notation: instead of $\mathbf{h}$ and H , which are standard in analysis, we use $\mathbf{p}$ and P , which are standard in optimization.

Theorem (Taylor's theorem for vector-variate functions)

First-order approximation: If $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is C^1 at $\mathbf{x} \in \mathbb{R}^n$, then for any nonzero $\mathbf{p} \in \mathbb{R}^n$,

$$f(\mathbf{x} + \mathbf{p}) = f(\mathbf{x}) + \nabla f(\mathbf{x} + \xi \mathbf{p})^\top \mathbf{p}$$

for some $\xi \in (0, 1)$.

Second order approximation: If $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is C^2 at $\mathbf{x} \in \mathbb{R}^n$, then for any nonzero $\mathbf{p} \in \mathbb{R}^n$,

In terms of ∇f :

$$\nabla f(\mathbf{x} + \mathbf{p}) = \nabla f(\mathbf{x}) + \int_0^1 \nabla^2 f(\mathbf{x} + t\mathbf{p}) \mathbf{p} dt$$

In terms of f :

$$f(\mathbf{x} + \mathbf{p}) = f(\mathbf{x}) + \nabla f(\mathbf{x})^\top \mathbf{p} + \frac{1}{2} \mathbf{p}^\top \nabla^2 f(\mathbf{x} + \xi \mathbf{p}) \mathbf{p}$$

for some $\xi \in (0, 1)$.

Theorem (Taylor's theorem for matrix-variate functions)

First-order approximation: If $f : \mathbb{R}^{m \times n} \rightarrow \mathbb{R}$ is C^1 at $X \in \mathbb{R}^{m \times n}$, then for any nonzero $P \in \mathbb{R}^{m \times n}$,

$$f(X + P) = f(X) + \text{tr}(\nabla f(X + \xi P)^\top P)$$

for some $\xi \in (0, 1)$.

Second order approximation: If $f : \mathbb{R}^{m \times n} \rightarrow \mathbb{R}$ is C^2 at $X \in \mathbb{R}^{m \times n}$, then for any nonzero $P \in \mathbb{R}^{m \times n}$, (a) in terms of ∇f ,

$$\nabla f(X + P) = \nabla f(X) + \int_0^1 [\nabla^2 f(X + tP)] (P) dt$$

In terms of f ,

$$f(X + P) = f(X) + \text{tr}(\nabla f(X)^\top P) + \frac{1}{2} \text{tr}(P^\top [\nabla^2 f(X + \xi P)] (P))$$

for some $\xi \in (0, 1)$.

The domain of f can be any open $\Omega \subseteq \mathbb{R}^n$ or $\Omega \subseteq \mathbb{R}^{m \times n}$.

The first order approximation is also called the mean value theorem.

The remainder for ∇f cannot be written in a differential form, i.e., in general there is no $\xi \in (0, 1)$ such that

$$\begin{aligned}\nabla f(\mathbf{x} + \mathbf{p}) &= \nabla f(\mathbf{x}) + \nabla^2 f(\mathbf{x} + \xi \mathbf{p}) \mathbf{p} \\ \nabla f(X + P) &= \nabla f(X) + \nabla^2 f(X + \xi P) P\end{aligned}$$

The integral in the second order approximation refers to the integration of a vector-valued or matrix-valued univariate function.

For $F : \mathbb{R} \rightarrow \mathbb{R}^n$ or $F : \mathbb{R} \rightarrow \mathbb{R}^{m \times n}$ given by

$$F(t) = \begin{bmatrix} f_1(t) \\ \vdots \\ f_n(t) \end{bmatrix} \quad \text{or} \quad F(t) = [f_{ij}(t)]_{i,j=1}^{m,n}$$

where f_i 's and f_{ij} 's are integrable on $[a, b]$, we define

$$\int_a^b F(t)dt = \begin{bmatrix} \int_a^b f_1(t)dt \\ \vdots \\ \int_a^b f_m(t)dt \end{bmatrix} \quad \text{or} \quad \int_a^b F(t)dt = \left[\int_a^b f_{ij}(t)dt \right]_{i,j=1}^{m,n}$$

Convex sets

Convex optimization problems are among the most important and prevalent minimization problems in recent years.

Many examples that we mentioned in earlier lectures are all convex optimization problems:

- linear programming (LP)
- quadratic programming (QP)
- second-order cone programming (SOCP)
- semidefinite programming (SDP)
- geometric programming (GP)

Again for simplicity, we state our results for $\mathcal{U} = \mathbb{R}^n$ but everything below applies to any vector space $\mathcal{U}$ such as $\mathbb{S}^n, \mathbb{R}^{m \times n}, \mathbb{R}^n \times \mathbb{S}^n$, etc.

We say that a set is convex if for any two points in the set, every point on the line segment connecting them is also in the set.

So for example, a ball is convex, but a donut or a crescent moon are not. (Draw.)

Definition (Line segment)

For any two points $\mathbf{x}, \mathbf{y}$ in $\mathbb{R}^n$, the line segment $[\mathbf{x}, \mathbf{y}]$ is the set of all points in the line between $\mathbf{x}$ and $\mathbf{y}$. Mathematically,

$$[\mathbf{x}, \mathbf{y}] = \{t\mathbf{x} + (1 - t)\mathbf{y} \mid t \in [0, 1]\}.$$

Definition (Convex set)

Let Ω be a subset of $\mathbb{R}^n$. We say Ω is convex iff for any $\mathbf{x}, \mathbf{y} \in \Omega$,

$$t\mathbf{x} + (1 - t)\mathbf{y} \in \Omega$$

for all $t \in [0, 1]$. That is, $[\mathbf{x}, \mathbf{y}] \subset \Omega$ for all $\mathbf{x}, \mathbf{y} \in \Omega$.

All of these are convex:

- open/closed balls
- convex cones (cones that contain their interiors – like filled ice cream cones)
- ellipsoids
- hyperplanes
- vector/affine subspaces (lines and planes, but potentially shifted away from the origin)
- simplices (generalization of triangles to higher dimensions)

Definition (Cone)

A subset C of $\mathbb{R}^n$ is said to be a cone iff for any $x \in C$ and $\alpha > 0$, we have $\alpha x \in C$.

$$[\mathbf{x}, \mathbf{y}] = \{t\mathbf{x} + (1-t)\mathbf{y} \mid t \in [0, 1]\}.$$

We can express any point on a line segment as a weighted average of the two endpoints.

For $t = 0$, we get $\mathbf{x}$ and for $t = 1$, we get $\mathbf{y}$. For $t = 0.5$, we get the midpoint.

Can we generalize this to express the connection between more than two points?

Yes, we can do this for any finite collection of points – we just want to make a weighted average of the points, where the weights are non-negative and sum to 1. (Draw.)

This is called a convex combination.

Definition (Convex combination)

A convex combination of points $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_k$ in $\mathbb{R}^n$ is

$$\alpha_1 \mathbf{x}_1 + \alpha_2 \mathbf{x}_2 + \dots + \alpha_k \mathbf{x}_k,$$

where $\alpha_1, \dots, \alpha_k \geq 0$ and $\alpha_1 + \dots + \alpha_k = 1$.

The **convex hull** is the generalization of the notion of a line segment that works more than two points: it is the set of all convex combinations of the points.

The convex hull contains all the points that are on line segments between any two points in the set, but it can also contain more than these points.

For example, let's take a set of three points in $\mathbb{R}^2$: (draw it)

$$S = \{(1, 1), (2, 2), (3, 1)\}$$

The convex hull of this set of three points is the triangle formed by the three points, plus all the points in the interior of the triangle. ##

Definition (Convex hull)

Let $S \subseteq \mathbb{R}^n$. The convex hull of S is the set

$$\text{conv}(S) := \{\alpha_1 \mathbf{x}_1 + \alpha_2 \mathbf{x}_2 + \cdots + \alpha_k \mathbf{x}_k \in \mathbb{R}^n : \mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_k \in S \\ \alpha_1, \dots, \alpha_k \geq 0, \alpha_1 + \cdots + \alpha_k = 1, k \in \mathbb{N}\}.$$

A set S is convex if and only if the convex hull of S is equal to S .

So in our triangle example, the set of three vertices is not convex, but the triangle including the interior is convex.

Exercise

Show that $\text{conv}(S)$ is the smallest convex set containing S and $\text{conv}(S) = S$ iff S is convex.

Let S, S_1, S_2 be convex sets. Convexity of sets is preserved under the following operations:

Intersection: $S_1 \cap S_2$ is convex. More generally if $S_i, i \in I$, are all convex sets, then

$$\bigcap_{i \in I} S_i$$

is convex (here the index set I can be infinite or uncountable).

What about the union?

Draw it.

Let $\alpha > 0$ and $\mathbf{a} \in \mathbb{R}^n$. Then

$$\alpha S := \{\alpha \mathbf{x} \in \mathbb{R}^n : \mathbf{x} \in S\}, \quad S + \mathbf{a} := \{\mathbf{x} + \mathbf{a} \in \mathbb{R}^n : \mathbf{x} \in S\}$$

are convex. More generally if $T : \mathbb{R}^m \rightarrow \mathbb{R}^n$ is a linear map, then the set

$$\{\mathbf{x} \in \mathbb{R}^m : T(\mathbf{x}) + \mathbf{a} \in S\}$$

is convex.

$S_1 + S_2 := \{\mathbf{x}_1 + \mathbf{x}_2 : \mathbf{x}_1 \in S_1, \mathbf{x}_2 \in S_2\}$, $S_1 \times S_2 := \{(\mathbf{x}_1, \mathbf{x}_2) : \mathbf{x}_1 \in S_1, \mathbf{x}_2 \in S_2\}$
are convex.

The set of positive semidefinite matrices is convex

Can we show that the set of positive semidefinite matrices is convex?

Yes, we can.

We'll start by picking a vector $\mathbf{y}$ and showing that the half-space of form $\{X \in \mathbb{S}^n : \mathbf{y}^\top X \mathbf{y} \geq 0\}$ is convex.

We will use the definition of a convex set in terms of line segments.

We will take any two matrices X_1 and X_2 in this half-space, and show that any point on the line segment connecting them is also in the half-space.

For any point on the line segment $X_t = tX_1 + (1 - t)X_2$ where $t \in [0, 1]$

We can evaluate: $\mathbf{y}^\top X_t \mathbf{y} = \mathbf{y}^\top (tX_1 + (1 - t)X_2) \mathbf{y}$

Using linearity: $= t\mathbf{y}^\top X_1 \mathbf{y} + (1 - t)\mathbf{y}^\top X_2 \mathbf{y}$

Since $\mathbf{y}^\top X_1 \mathbf{y} \geq 0$ and $\mathbf{y}^\top X_2 \mathbf{y} \geq 0$, and $t, 1 - t \geq 0$, this sum must be non-negative, so X_t is also in the half-space.

Now we wanted to show that the set of positive semidefinite matrices is convex.

What we have shown above is for a specific vector $\mathbf{y}$, but we want this to be true in general:

$$\mathbb{S}_+^n = \{X \in \mathbb{S}^n : \mathbf{y}^\top X \mathbf{y} \geq 0 \text{ for all } \mathbf{y} \in \mathbb{R}^n\}$$

The set of positive semidefinite matrices is the *intersection* of all half-spaces of form $\{X \in \mathbb{S}^n : \mathbf{y}^\top X \mathbf{y} \geq 0\}$ for all $\mathbf{y} \in \mathbb{R}^n$.

$$\mathbb{S}_+^n = \bigcap_{\mathbf{y} \in \mathbb{R}^n} \{X \in \mathbb{S}^n : \mathbf{y}^\top X \mathbf{y} \geq 0\}$$

Since the intersection of convex sets is convex, $\mathbb{S}_+^n$ is convex.

Convex functions

Intuitive definition of a convex function

A **function is convex** if the set of all points on or above the graph of the function is a convex set.

Specifically, for any two points x, y in the domain of the function, the line segment connecting them lies on or above the graph of the function.

For example, the function $f(x) = x^2$ is convex because the set of all points above the graph of the function is a convex set.

But the function $f(x) = -x^2$ is not convex because the set of all points above the graph of the function is not a convex set.

We can think of a convex function as a function that always curves upward, like a bowl, or is flat, but never curves downward.

What if the domain is not convex?

If we want to test that the line segment connecting two points lies on or above the graph of the function, we need the function to be defined everywhere between the two points.

(Draw counter-example, with a function that isn't defined between 0 and 1)

This means that the domain of a convex function is always a convex set (otherwise the definition makes no sense).

Formal definition of a convex function

A function $f : \Omega \rightarrow \mathbb{R}$ is convex if for all $\mathbf{x}, \mathbf{y} \in \Omega$ and $t \in [0, 1]$,

$$f(t\mathbf{x} + (1-t)\mathbf{y}) \leq tf(\mathbf{x}) + (1-t)f(\mathbf{y})$$

We need to evaluate the function at point $t\mathbf{x} + (1-t)\mathbf{y}$ for any $\mathbf{x}, \mathbf{y}$ in the domain and any $t \in [0, 1]$.

If the domain weren't convex, there would exist some $\mathbf{x}, \mathbf{y}$ in the domain and some $t \in [0, 1]$ where $t\mathbf{x} + (1-t)\mathbf{y}$ falls outside the domain. Then we couldn't evaluate f at this point, making the inequality impossible to check.

Definition (Convex function)

Let $\Omega \subset \mathbb{R}^n$ be a convex set. Then $f : \Omega \rightarrow \mathbb{R}$ is a convex function iff for all $\mathbf{x}, \mathbf{y} \in \Omega$ and $t \in [0, 1]$,

$$f(t\mathbf{x} + (1-t)\mathbf{y}) \leq tf(\mathbf{x}) + (1-t)f(\mathbf{y})$$

Theorem

Let $f : \Omega \rightarrow \mathbb{R}$ be convex. Then every local minimizer is a global minimizer.

Furthermore, if $f \in C^1(\Omega)$, then $\nabla f(\mathbf{x}_) = \mathbf{0}$ implies $\mathbf{x}_*$ is a global minimizer, i.e. a stationary point is automatically a global minimizer.*

In other words, for convex optimization problems, we can do global optimization!

In fact the current wisdom is that if we are given a non-convex optimization problem, the best way to approach it is to first approximate the problem by a convex optimization problem - this step is usually call convex relaxation - and then solve its convex relaxation instead.

Conditions for convexity

To check if a function f is convex over a convex domain Ω , one can use any of the following equivalent characterizations of convexity:

$f : \Omega \rightarrow \mathbb{R}$ is convex if and only if $f(t\mathbf{x} + (1-t)\mathbf{y}) \leq tf(\mathbf{x}) + (1-t)f(\mathbf{y})$. (Definition)

$f \in C^1(\Omega)$ is convex if and only if $f(\mathbf{y}) \geq f(\mathbf{x}) + \nabla f(\mathbf{x})^\top (\mathbf{y} - \mathbf{x})$ for all $\mathbf{x}, \mathbf{y} \in \Omega$.
("First order condition")

$f \in C^2(\Omega)$ is convex if and only if $\nabla^2 f(\mathbf{x}) \succeq 0$ for all $\mathbf{x} \in \Omega$. ("Second order condition")

From the second order condition, it is immediately clear why stationary points are automatically global minimizers for convex functions - the positive semidefiniteness of the Hessian is always guaranteed for a convex function.

There are also zeroth order conditions that are less commonly used.

$f \in C^0(\Omega)$ is convex if and only if

$$\int_0^1 f(\mathbf{x} + t(\mathbf{y} - \mathbf{x}))dt \leq \frac{f(\mathbf{x}) + f(\mathbf{y})}{2}$$

for all $\mathbf{x}, \mathbf{y} \in \Omega$.

(Draw it.)

This is a generalization of the first order condition to integrals. Intuitively, it says that the function is convex if and only if the area under the function is less than or equal to the area of the rectangle whose height is the average of the function values at the endpoints.

For univariate functions,

note that $\Omega \subseteq \mathbb{R}$ is convex iff Ω is an interval (a, b) , $[a, b)$, $(a, b]$, or $[a, b]$;

$f \in C^0(\Omega)$ is convex if and only if

$$f\left(\frac{x+y}{2}\right) \leq \frac{f(x) + f(y)}{2}$$

for all $x, y \in \Omega$;

if Ω is open, i.e., $\Omega = (a, b)$, then any convex $f : (a, b) \rightarrow \mathbb{R}$ must be continuous;

not true if Ω is not open, e.g., $f : (0, 1] \rightarrow \mathbb{R}$, $f(x) = x$ for $x \in (0, 1)$, $f(1) = 2$.

Theorem (First order condition for convexity)

Let $f \in C^1(\Omega)$, where Ω is convex. Then f is a convex function iff

$$f(\mathbf{y}) \geq f(\mathbf{x}) + \nabla f(\mathbf{x})^\top (\mathbf{y} - \mathbf{x})$$

for all $\mathbf{x}, \mathbf{y} \in \Omega$.

Proof of the first order condition

Proof. ($\Rightarrow$) Suppose f is convex, and let $t \in (0, 1]$. Then

$$f(\mathbf{x} + t(\mathbf{y} - \mathbf{x})) = f(t\mathbf{y} + (1 - t)\mathbf{x}) \leq (1 - t)f(\mathbf{x}) + tf(\mathbf{y})$$

Dividing by $t \neq 0$ and re-arranging,

$$f(\mathbf{y}) \geq f(\mathbf{x}) + \frac{f(\mathbf{x} + t(\mathbf{y} - \mathbf{x})) - f(\mathbf{x})}{t}$$

Letting $t \rightarrow 0^+$,

$$f(\mathbf{y}) \geq f(\mathbf{x}) + \lim_{t \rightarrow 0^+} \frac{f(\mathbf{x} + t(\mathbf{y} - \mathbf{x})) - f(\mathbf{x})}{t} = f(\mathbf{x}) + \nabla f(\mathbf{x})^\top (\mathbf{y} - \mathbf{x})$$

as desired.

($\Rightarrow$) Now suppose $(\star)$ holds, and fix $\mathbf{x}, \mathbf{y} \in \Omega$. Let $\mathbf{z} = t\mathbf{x} + (1-t)\mathbf{y}$. Applying $(\star)$ twice,

$$f(\mathbf{x}) \geq f(\mathbf{z}) + \nabla f(\mathbf{z})^\top (\mathbf{x} - \mathbf{z})$$

$$f(\mathbf{y}) \geq f(\mathbf{z}) + \nabla f(\mathbf{z})^\top (\mathbf{y} - \mathbf{z})$$

Multiplying the first equation by t and second equation by $(1-t)$ and adding,

$$tf(\mathbf{x}) + (1-t)f(\mathbf{y}) \geq f(\mathbf{z}) = f(t\mathbf{x} + (1-t)\mathbf{y})$$

There is a related notion of concavity:

Definition (Concave function)

Let $\Omega \subset \mathbb{R}^n$ be a convex set. Then $f : \Omega \rightarrow \mathbb{R}$ is a concave function iff for all $\mathbf{x}, \mathbf{y} \in \Omega$ and $t \in [0, 1]$,

$$f(t\mathbf{x} + (1-t)\mathbf{y}) \geq tf(\mathbf{x}) + (1-t)f(\mathbf{y})$$

Note that a concave function must also be defined on a convex set.

We will not use concavity since it is essentially equivalent to convexity: It is easy to see that a function f is concave if and only if $-f$ is convex.

Theorem (Second order condition for convexity)

Let $f \in C^2(\Omega)$, where Ω is convex. Then f is convex if and only if $\nabla^2 f(\mathbf{x}) \succeq 0$ for all $\mathbf{x} \in \Omega$.

Proof. For simplicity, we prove this in the univariate case only.

($\Rightarrow$) Fix $x, y \in \Omega$, with $x \neq y$. If f is convex, the first order conditions hold:

$$f(y) \geq f(x) + f'(x)(y - x)$$

$$f(x) \geq f(y) + f'(y)(x - y)$$

Subtracting $f(x)$ from the first inequality and $f(y)$ from the second inequality,

$$f(y) - f(x) \geq f'(x)(y - x)$$

$$-(f(y) - f(x)) \geq -f'(y)(y - x)$$

Adding the inequalities,

$$\begin{aligned}0 &\geq (f'(x) - f'(y))(y - x) \\ (f'(y) - f'(x))(y - x) &\geq 0\end{aligned}$$

Dividing by $(y - x)^2 \neq 0$,

$$\frac{f'(y) - f'(x)}{y - x} \geq 0$$

Letting $y \rightarrow x$, we have $f''(x) \geq 0$, as desired.

($\Leftarrow$) Suppose $f''(z) \geq 0$ for all $z \in \Omega$, and let $x, y \in \Omega$ with $x < y$. Then

$$\begin{aligned} 0 &\leq \int_x^y f''(z)(y-z)dz \\ &= f'(z)(y-z) \Big|_{z=x}^{z=y} + \int_x^y f'(z)dz \\ &= -f'(x)(y-x) + (f(y) - f(x)) \end{aligned}$$

Thus

$$f(y) \geq f(x) + f'(x)(y-x)$$

By the first order condition, f is convex.

Everything we discussed above extends to matrix variate functions easily.

For the second order condition we need the following interpretation of positive (semi)definiteness for linear maps.

Definition (Positive semidefinite/definite)

A linear map $T : \mathbb{U} \rightarrow \mathbb{U}$ is said to be positive semidefinite if

$$\langle h, T(h) \rangle \geq 0 \quad \text{for all } h \in \mathbb{U};$$

and positive definite if

$$\langle h, T(h) \rangle > 0 \quad \text{for all } 0 \neq h \in \mathbb{U}.$$

In other words, positive semi(definiteness is well-defined for any linear map from an inner product space to itself. Check that these reduce to the usual definitions for positive (semi)definite matrices when $\mathbb{U} = \mathbb{R}^n$.

Example: Matrix variate functions

For $\mathbb{U} = \mathbb{R}^{m \times n}$ and $f : \Omega \subseteq \mathbb{R}^{m \times n} \rightarrow \mathbb{R}$.

Its Hessian is a linear map $\nabla^2 f(X) : \mathbb{R}^{m \times n} \rightarrow \mathbb{R}^{m \times n}$.

we see that it is positive semidefinite if

$$\text{tr}(H^\top [\nabla^2 f(X)](H)) \geq 0 \quad \text{for all } H \in \mathbb{R}^{m \times n}$$

and positive definite if

$$\text{tr}(H^\top [\nabla^2 f(X)](H)) > 0 \quad \text{for all } 0 \neq H \in \mathbb{R}^{m \times n}$$

If these hold for all $X \in \Omega$, then these are second order conditions that characterize the convexity and strict convexity (see next section) of f .